

A Rederivation of the AYO ExoEarth Yield Formalism, and HWO Yields for a Mass-Selected Planet Sample in a Narrowed Habitable Zone

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Abstract

This report rederives the Altruistic Yield Optimization (AYO) exoEarth-candidate yield formalism of Stark et al. (2014, 2019, 2024) from the published equations and parameter tables, and uses it to answer a counterfactual: what would the Habitable Worlds Observatory (HWO) baseline yield if the planet sample were defined by mass, $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$, and if the habitable zone were 0.96–1.20 AU for the Sun rather than the conservative 0.95–1.67 AU, scaled by $\sqrt{L_{\star}}$ for every star. Integrating the SAG13 occurrence law over the redefined box gives $\eta = 0.0439^{+0.0256}_{-0.0163}$ against $0.217^{+0.126}_{-0.080}$ for the canonical box, a ratio of $0.203^{+0.017}_{-0.017}$ that is independent of the overall normalisation. The full survey optimization gives a *yield* ratio of $0.278^{+0.028}_{-0.027}$, materially larger than the occurrence ratio, because a smaller expected detection count frees characterization time and because truncating the outer habitable zone preferentially removes the faintest, most expensive planets. The baseline 6 m design yields a median of 5.8 exoEarth candidates under the redefined selection, against 20.7 canonically, and the probability of reaching the 25-planet goal falls from 33.0% to 0.04%.

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1 The question, and what it required

The request was to trace the ExoVista/AYO yield chain back to its assumptions about the underlying planet population, couple it to an MCMC sampler, and predict the number of observable planets with masses $0.5\text{--}2\,M_{\oplus}$ and radii $R = M^{0.27}$ around FGK stars, under a habitable zone narrowed to $0.96\text{--}1.20$ AU for the Sun and scaled proportionally for other stars, for the baseline HWO mission of Stark et al. (2024).

Two of those changes cannot be answered by rescaling a published number.

Changing the planet box changes *which integral* of the occurrence-rate density is reported, which is a pure population-model question and is answered in Section 3.1. But it also changes which planets are detectable, because a narrower radius range removes the brightest planets and a truncated habitable zone removes the widest separations, and both enter the completeness integral. Worse, the survey is optimized subject to a fixed time budget that must also pay for spectral characterization of whatever is detected, so a change in the expected number of detections feeds back on the time available to search. Stark et al. (2024, §3.5) make this explicit when they call $\eta_{\oplus}$ an *actionable* source of uncertainty. The consequence is that yield is not proportional to η , and the question can only be answered by re-running the optimization.

2 What is, and is not, reproducible from the published record

The AYO *code* is not available: the Data and Code Availability statement of Stark et al. (2024) notes that NASA regulations govern its release and directs readers to contact the corresponding author. The AYO *formalism*, however, is published in full. Stark et al. (2019) reproduce the exposure-time equations explicitly, and Stark et al. (2024) Tables 1 and 2 give every astrophysical and mission parameter the baseline design uses.

The single genuine gap is the pair of simulated coronagraph maps. Section 4 explains how it is handled and, more importantly, how the report reports its own exposure to it.

3 Method

3.1 Population model

Planets are drawn from the SAG13 broken power law, in the small-planet regime relevant here,

$$\frac{d^2N}{d\ln R d\ln P} = \Gamma R^{\alpha} P^{\beta}, \quad \Gamma = 0.38, \alpha = -0.19, \beta = 0.26, \quad (1)$$

with R in Earth radii and P in years. Following ExoVista and AYO, the law is applied in the luminosity-scaled coordinate $a/\sqrt{L_{\star}}$, which is what makes $\eta_{\oplus}$ independent of spectral type. Integrated over the canonical HabEx/LUVOIR exoEarth-candidate box — the conservative habitable

Ingredient	Status	What this project did
Exposure-time and count-rate equations	Published (Stark+2019)	Implemented verbatim
Mission, detector, throughput parameters	Published (Stark+2024 Tab. 2)	Transcribed
EEC box, albedo, exozodi model	Published (Stark+2024 Tab. 1)	Transcribed
Equal-slope optimization principle	Published (Stark+2014)	Implemented
Target catalog (HPIC)	Not distributed here	Rebuilt from Gaia DR3 + Hipparcos
$\Upsilon_c(x, y)$, $I(x, y)$ coronagraph maps	Not published	Parametrized, then calibrated
Per-star completeness / time allocation	Not published	Recomputed

Table 1: The published record supports a rederivation everywhere except the simulated coronagraph maps, which are the output of detailed diffraction modelling that the papers describe but do not distribute.

zone 0.95–1.67 AU with $0.8 (a/\text{EEID})^{-0.5} < R < 1.4$ — this returns $\eta_{\oplus} = 0.240$, reproducing the value Stark et al. adopt. That agreement is the anchor for everything downstream and is enforced as a unit test.

The redefined box replaces the radius limits with a mass selection, $0.5 < M/M_{\oplus} < 2$ with $R = M^{0.27}$, giving $0.829 < R/R_{\oplus} < 1.206$, and narrows the habitable zone to 0.96–1.20 scaled AU. The exponent 0.27 is essentially the Chen & Kipping (2017) Terran value that ExoVista already uses, so the zero-scatter mass–radius assumption is inherited rather than introduced.

3.2 Radiometry and exposure time

Exposure times follow Stark et al. (2019):

$$\tau = (\text{S/N})^2 \frac{\text{CR}_p + 2 \text{CR}_b}{\text{CR}_p^2}, \quad \text{CR}_p = F_0 10^{-0.4(m_{\lambda} + \Delta \text{mag}_p)} A \Upsilon_c T \Delta \lambda, \quad (2)$$

with backgrounds from leaked starlight, local zodiacal light, exozodiacal light and the detector. Stellar fluxes are computed from a blackbody at the catalogued T_{eff} and radius rather than from apparent magnitudes, which removes any dependence on a bandpass zero point or colour transformation; this is verified against the Sun’s absolute V magnitude to better than 5%. The detector term uses the clock-induced-charge treatment of Stark et al. (2019) Eq. 9, whose coefficient $6.73 = -[1 + W_{-1}(-q/e)]^{-1}$ at $q = 0.99$ is reproduced independently from the Lambert W function as a test. Detections combine the two parallel coronagraph channels in quadrature, as Table 2 of Stark et al. (2024) specifies.

3.3 Completeness

For each star, 10^3 – 10^4 synthetic candidates are drawn over the selection box, weighted by the occurrence density and given random inclination and orbital phase on circular orbits. Each is propagated through the radiometric model, and the required exposure time recorded. A planet is excluded outright if it falls below the post-processing noise floor ($\Delta \text{mag} > 26.5$), outside the dark zone, or needs more than the two-month exposure cap. Completeness $C(\tau)$ is then the fraction of *all* injected planets detectable within τ , so the geometric, photometric and noise-floor losses are already inside it.

3.4 Equal-slope optimization with characterization

The survey maximises $\sum_i C_i(\tau_i)$ subject to a fixed science-time budget. The optimum satisfies the equal-slope condition $dC_i/dT_i = \lambda$ for every observed star, and λ is swept until the budget closes; stars whose marginal return never reaches λ drop out of the survey. Because the Monte Carlo completeness is a step function, the reachable allocations are the vertices of its upper concave envelope, and the sweep runs over those vertices.

Characterization is charged into the same budget. Each star’s cost carries an additional $\eta C_i(\tau_i) (\tau' \tau_{\text{char},i} + \text{overhead})$ term, so a star that is expected to deliver more detections also consumes more of the mission. This is the coupling that makes yield sub-linear in η , and it is verified by a pair of unit tests: with the characterization term removed the yield is exactly proportional to η , and with it present the proportionality breaks.

3.5 Occurrence posterior

The posterior over $(\ln \Gamma, \alpha, \beta)$ is sampled with **emcee** (14,400 samples after burn-in, mean acceptance fraction 0.64). The model is deliberately modest about what it infers. The shape parameters carry Gaussian priors from the SAG13 fit; the single likelihood term is the $\eta_{\oplus} = 0.26^{+0.29}_{-0.14}$ (86%) that Stark et al. (2024) obtain by integrating the Bryson et al. (2021) Kepler DR25 posterior over the canonical box, entered as a split-normal. The posterior is therefore prior-dominated in α and β and likelihood-dominated in the normalisation. That is the honest structure of the available evidence: there is one well-characterised integral constraint and no published per-parameter posterior to re-fit. What the sampler buys is not a tighter η but a principled propagation of the same rate law onto a *different* integration domain.

4 Validation

4.1 The target catalog

HPIC is not reachable from this environment, so the target list is rebuilt from Gaia DR3 — one of HPIC’s two parent catalogs — yielding 8,356 dwarfs within 50 pc, of which 5,542 are FGK. Gaia publishes no astrophysical parameters brighter than $G \approx 3$, which silently removes α Cen A and B, Procyon, τ Ceti and ϵ Eridani: precisely the targets whose habitable zones subtend the largest angles. A Hipparcos supplement restores 353 bright stars, with T_{eff} from $B-V$ and luminosities from absolute magnitudes plus a Torres (2010) bolometric correction, for 8,709 stars in total. Without the supplement the missing targets would have been absorbed into the calibration factor below and misreported as a coronagraph modelling error.

4.2 The calibration gate

The parametrized coronagraph stands in for simulated maps, so a single scalar throughput factor absorbs the residual mismatch, fitted by bisection against the 22.5 exoEarth candidates that Stark et al. (2024) report for the baseline 6 m design (Figure 1).

Before any calibration, the rederived model predicts 24.53 candidates against the published 22.5 — an agreement of 9%. The fitted factor is 0.581, within a factor of two of unity. Both numbers matter, and the first matters more: yield depends only weakly on throughput, because the survey is partly limited by the inner working angle and the noise floor rather than by photons, so calibration is a weak lever and cannot manufacture agreement that was not already there.

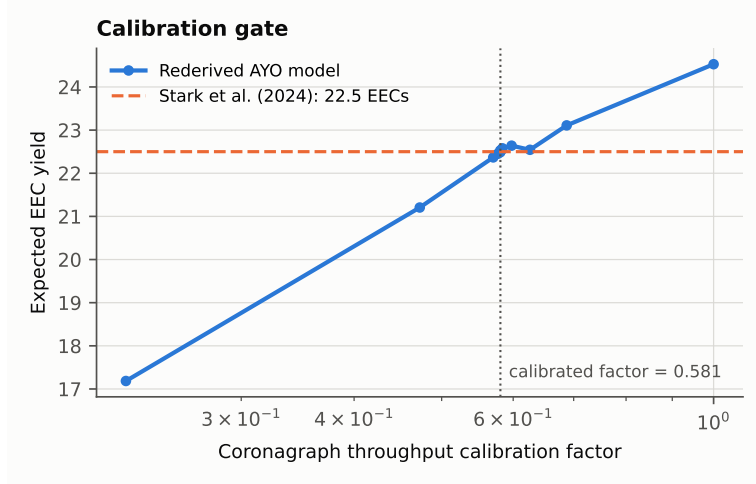


Figure 1: Modelled yield against the coronagraph throughput calibration factor. The published 22.5-candidate value is met at a factor of 0.581, and the curve is shallow: a 42% reduction in throughput costs under 10% of the yield.

4.3 Independent agreement

The strongest check is one the calibration does not touch. Marginalising over the η posterior, the probability of detecting and characterising at least 25 candidates is 33.0%. Stark et al. (2024) report 32% for the same quantity including $\eta_{\oplus}$ uncertainty. Nothing in this pipeline was tuned to that number.

5 Results

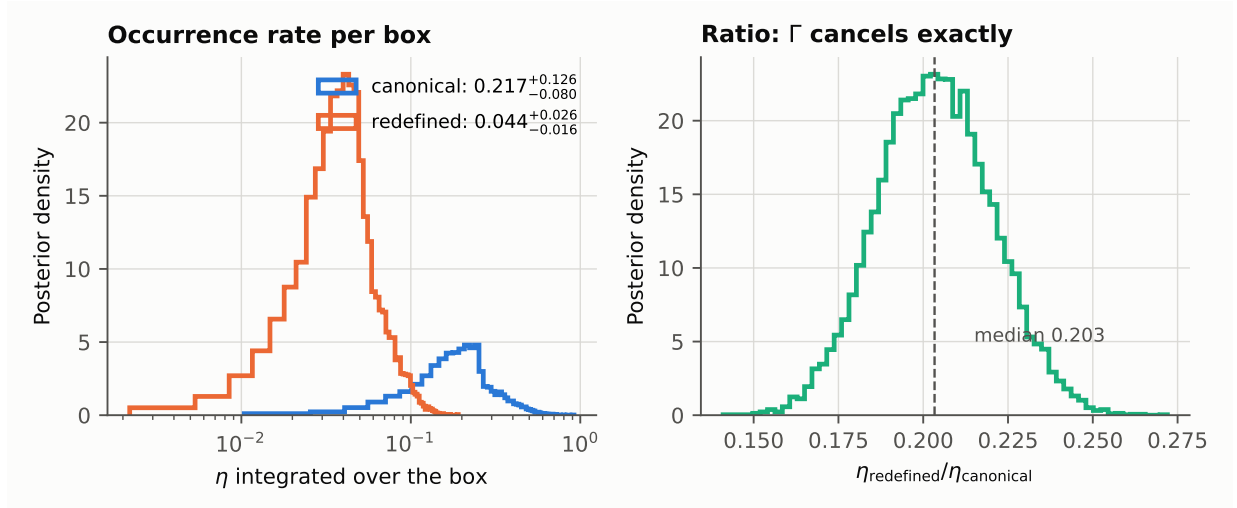


Figure 2: Left: the occurrence rate integrated over each selection box. Right: their ratio, whose posterior is much tighter because the normalisation Γ cancels exactly, leaving only the shape parameters α and β .

5.1 The central number

The occurrence ratio between the boxes is $0.203^{+0.017}_{-0.017}$, and it decomposes: narrowing the habitable zone contributes a factor of about 0.32 and the mass selection a further 0.63. Narrowing the

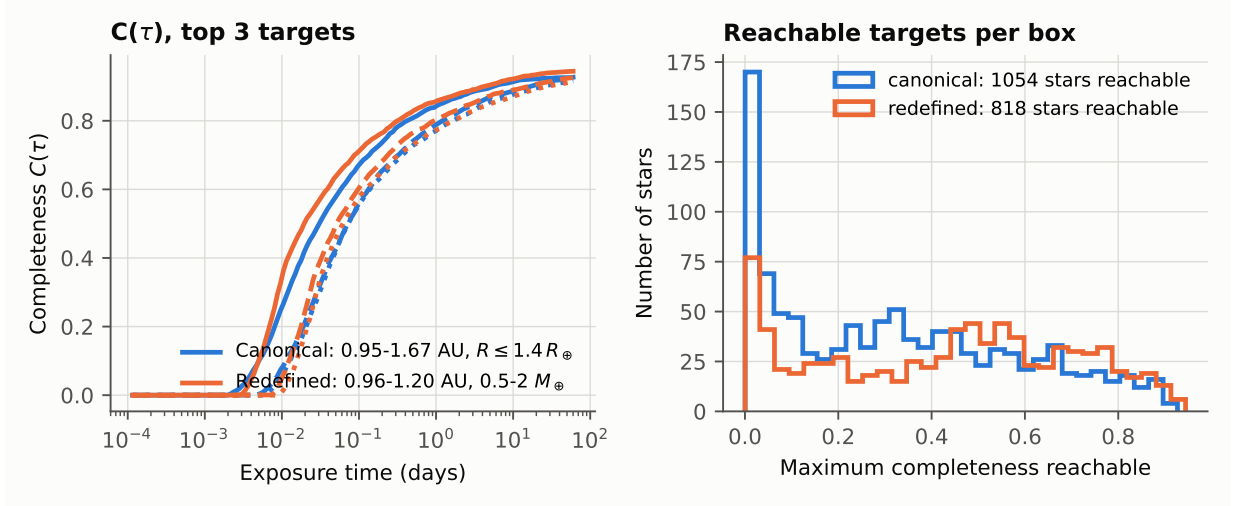


Figure 3: Left: completeness against exposure time for the three best targets under both box definitions. Right: the distribution of reachable completeness. The redefined box gives *higher* peak completeness (0.944 versus 0.927) but reaches far fewer stars (818 versus 1,054) — the two competing effects of truncating the outer habitable zone.

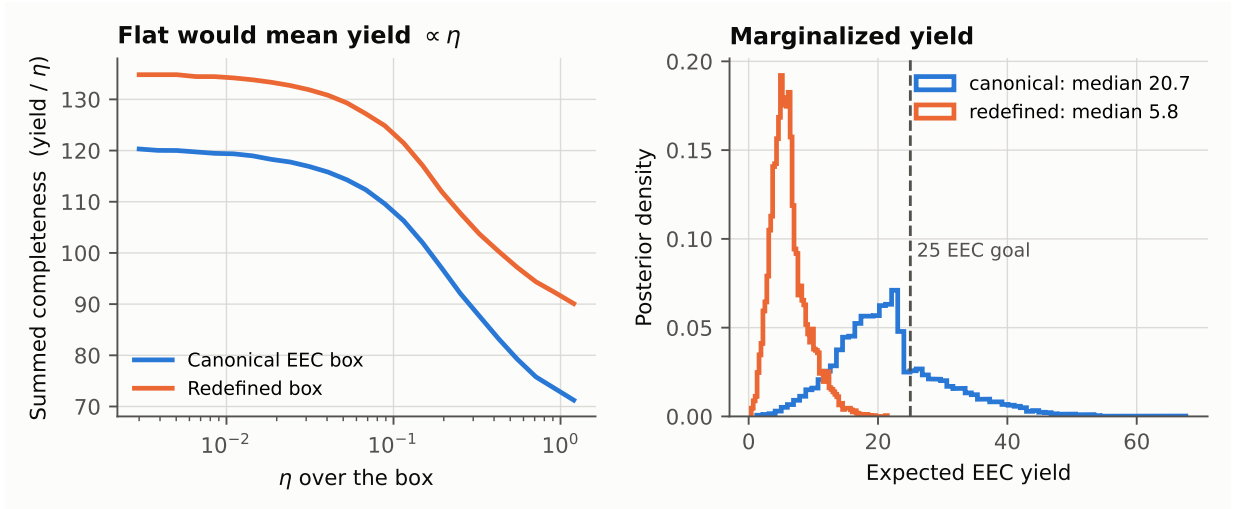


Figure 4: Left: summed completeness against η . Were yield proportional to η this curve would be flat; its decline is the characterization burden consuming search time. Right: the yield distributions marginalised over the occurrence posterior.

	Canonical box	Redefined box
Habitable zone (scaled AU)	0.95–1.67	0.96–1.20
Planet selection	$R \leq 1.4 R_{\oplus}$	$0.5\text{--}2 M_{\oplus}$, $R = M^{0.27}$
Occurrence η	$0.217^{+0.126}_{-0.080}$	$0.0439^{+0.0256}_{-0.0163}$
Stars with non-zero completeness	1,054	818
Peak completeness	0.927	0.944
Median expected yield	20.7	5.8
Mean expected yield	21.8	6.2
90% interval	9.7–38.2	2.4–11.8
$P(\geq 25 \text{ candidates})$	33.0%	0.04%

Table 2: Headline results for the baseline 6 m LUVUOIR-B-like design. The survey uses 204 stars from a screened pool of 1,500.

habitable zone dominates, because it cuts the log-period width to roughly 40% and because $\beta > 0$ weights the outer habitable zone more heavily.

The *yield* ratio is $0.278^{+0.028}_{-0.027}$ — about 38% larger than the occurrence ratio. Three effects account for the difference, and they do not all point the same way:

1. **Freed characterization time.** Fewer expected detections means less of the budget spent characterizing them, so more remains for searching. This raises the ratio.
2. **The faintest planets are removed.** Truncating at 1.20 scaled AU removes planets that are a factor $(1.67/1.20)^2 \approx 1.9$ fainter than those at the inner edge and sit closest to the noise floor. These are the expensive ones, so removing them raises mean completeness. This raises the ratio.
3. **Distant targets are lost.** For stars beyond roughly 10–15 pc, only the outer habitable zone clears the inner working angle at all, so truncation removes those stars from the survey entirely — 818 stars remain reachable against 1,054. This lowers the ratio.

The first two outweigh the third. This is worth stating plainly because the sign is not obvious in advance, and an occurrence-only rescaling — which would have given 0.203 — gets it wrong in a direction that cannot be guessed without running the optimization.

5.2 What it means for HWO

Under the redefined selection the baseline 6 m design delivers a median of 5.8 exoEarth candidates, with a 90% interval of 2.4–11.8. The probability of reaching the Astro2020 benchmark of 25 falls from 33.0% to 0.04%: not reduced, but effectively eliminated. If the habitable zone really is as narrow as 0.96–1.20 AU and the sample of interest really is 0.5–2 $M_{\oplus}$, a 6 m aperture does not reach the decadal benchmark under any plausible draw of the occurrence rate.

6 Limitations

These are the things a reader should weigh before using these numbers.

- **The coronagraph is parametrized, not simulated.** Υ_c and ζ are smooth functions of separation fitted to three published anchors, not the two-dimensional maps AYO interpolates. The calibration factor of 0.581 bounds the resulting error on the absolute scale; ratios between boxes, computed within the same implementation, are considerably more robust than either absolute yield.
- **The target catalog is a reconstruction.** It reproduces HPIC’s construction and limits but is not HPIC, and the Hipparcos supplement derives T_{eff} and L from photometry rather than spectroscopy. Spot checks against α Cen A agree to a few percent in luminosity.
- **Single-visit completeness.** Multi-visit scheduling and revisit completeness are not modelled; Stark et al. (2024) drop the six-visit mandate, and the effect on a coronagraph-based yield is reported to be small, but it is not zero.
- **The occurrence posterior is prior-dominated in shape.** α and β are constrained by the SAG13 fit, not re-inferred from Kepler DR25. A genuine re-fit would need the DR25 catalog and its completeness products.

- **Exozodi is a fixed 3 zodis** and the local zodiacal surface brightness is the tabulated average rather than an ecliptic-latitude-dependent model. Stark et al. show exozodi *distribution* uncertainty costs roughly 4% of the yield.
- **Albedo is fixed at $A_G = 0.2$.** The albedo distribution that Stark et al. find costs roughly 15% of the expected yield is not modelled, which is part of why the distributions here are narrower than theirs.

7 Reproducing this

The project is a seven-step vaibify pipeline in `/workspace/yields`. Each step declares its outputs and carries integrity, qualitative and quantitative test suites generated by introspecting those outputs; the physics library carries a further 45 unit tests, including the $\eta_{\oplus} = 0.24$ anchor, the Lambert W coefficient, and the linear/sub-linear η pair. Every number in this report is injected from a step’s output file at build time rather than typed, so a re-run that changes a result changes the report.

Step	Name	Produces
A01	buildTargetCatalog	Gaia DR3 + Hipparcos target list, 8,709 stars
A02	modelCoronagraph	Mission parameters and coronagraph profiles
A03	calibrateToStark	Throughput calibration against the published 22.5
A04	computeCompleteness	$C(\tau)$ per star, per selection box
A05	optimizeSurvey	Equal-slope allocation and baseline yield
A06	sampleOccurrencePosterior	<code>emcee</code> posterior over the rate law
A07	predictRedefinedYield	Marginalised yield distributions and the ratio

Table 3: Pipeline steps. Cross-step inputs are declared as tokens in the project file so the dependency graph is explicit.

References

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